Module 1 - DDCA

Important Questions - Clean Version

1. Computer Basics & RISC-V

1. Explain the Input–Process–Store–Output cycle with an example.
2. Differentiate between hardware and software with examples.
3. Explain the role of the CPU.
4. Explain the fetch–decode–execute cycle.
5. What is Instruction Set Architecture (ISA)? Why is it important?
6. Compare RISC and CISC.
7. Explain RISC-V and its advantages.
8. Differentiate RISC-V, RIPES, and Logisim.
9. Explain ALU, registers, Program Counter, Control Unit, instruction memory, and data memory.

2. Digital Systems & Number Systems

1. Why do computers use binary numbers?
2. Explain digital abstraction and noise immunity.
3. Define bit and byte.
4. Explain decimal, binary, octal, and hexadecimal number systems.
5. Convert 25, 45, and 237 from decimal to binary.
6. Convert 10101, 111000, and 1101011 from binary to decimal.
7. Convert 10101111 and 11110000 from binary to hexadecimal.
8. Convert AF and 3C from hexadecimal to binary.
9. Explain why hexadecimal is a convenient shorthand for binary.

3. Signed Numbers & Complements

1. Explain unsigned integer representation and its n-bit range.
2. Why are signed numbers needed?
3. Explain sign-magnitude representation.
4. Explain 1's-complement representation.
5. Explain 2's-complement representation.
6. Find the 1's complement of 10001, 111000, and 101101.
7. Find the 2's complement of 0101, 1010, and 001011.
8. Represent -5 using 8-bit two's complement.
9. Explain subtraction using two's-complement addition.

4. Binary Arithmetic

1. Explain the rules of binary addition.
2. Solve 1010 + 0111.
3. Solve 1110 + 1011.
4. Explain the rules of binary subtraction.
5. Solve 111000 - 100111.
6. Solve 101101 - 011011.
7. Solve 111000 - 100111 using the 1's-complement method.
8. Solve binary subtraction using the 2's-complement method.

5. Overflow

1. What is overflow? Explain with an example.
2. Explain unsigned overflow.
3. Explain signed overflow in two's complement.
4. Check overflow in 0111 + 0001.
5. Why can adding two positive signed numbers produce a negative result?

6. Logic Gates

1. What is a logic gate? Why is it important?
2. Write truth tables for AND, OR, NOT, NAND, NOR, XOR, and XNOR.
3. Explain AND with truth table and real-life application.
4. Explain OR with truth table and real-life application.
5. Explain NOT with truth table.
6. Explain NAND and NOR.
7. Compare XOR and XNOR.
8. Write Boolean expressions for the basic logic gates.
9. Explain how gates are combined to build larger digital circuits.

7. Boolean Algebra

1. Define Boolean algebra.
2. Explain Boolean variables and AND, OR, NOT operations.
3. State and explain Identity, Null, Idempotent, Complement, Involution, and Distributive laws.
4. State and explain De Morgan's laws.
5. Simplify Boolean expressions using Boolean laws.
6. Simplify A + A.B.
7. Simplify A(A + B).

8. Truth Table to SOP to Circuit

1. Explain how to convert a truth table into a Boolean expression.
2. What is Sum of Products (SOP) form?
3. Convert a given truth table into SOP.
4. Convert a Boolean expression into a logic-gate circuit.
5. For F = A'B + AB, give the gate-level implementation.

9. NAND Universality

1. What is a universal gate?
2. Why is NAND a universal gate?
3. Implement NOT using only NAND.
4. Implement AND using only NAND.
5. Implement OR using only NAND.
6. Explain how any Boolean function can be implemented using NAND.
7. Compare NAND and NOR as universal gates.

10. Tools & Practical Questions

1. What is Logisim Evolution and why is it used?
2. What is RIPES and why is it used?

3. Differentiate Logisim and RIPES.
4. Explain how to test a logic gate in Logisim.
5. Explain the basic simulation workflow.

Highest-Priority Questions

1. Explain CPU and the fetch-decode-execute cycle.
2. Explain RISC-V and ISA.
3. Practice binary-decimal-hexadecimal conversions.
4. Explain 1's complement and 2's complement.
5. Solve binary addition and subtraction.
6. Solve subtraction using 1's complement.
7. Explain overflow.
8. Learn truth tables for all logic gates.
9. Explain Boolean laws and De Morgan's laws.
10. Convert truth tables into SOP.
11. Convert Boolean expressions into circuits.
12. Explain NAND as a universal gate.
13. Differentiate Logisim and RIPES.